HitPredict: Predicting Billboard Hits Using Spotify Data

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1. Problem Statement

The popularity of a song depends on several factors, including its musical characteristics and existing popularity. With the availability of Spotify audio features, machine learning can be used to investigate whether these characteristics can help identify songs that become commercially successful.

The objective of **HitPredict** is to predict whether a song is a **Billboard Hot 100 hit** using Spotify-related song features. This is formulated as a binary classification problem:

- **1 → Billboard Hot 100 Hit**
- **0 → Non-Hit**

The main goal of the project is to preprocess song data, construct a suitable classification dataset, train different machine learning models, and compare their performance in predicting the hit class.

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2. Dataset Details

Two datasets were used to construct the final dataset.

The first dataset, **Billboard_Hot100_Songs_Spotify_1946-2022**, contains Billboard Hot 100 songs along with Spotify information and audio features. Since these songs are known Billboard Hot 100 entries, they were assigned the label **hit = 1**.

The second dataset, **playlist_2010to2023**, contains Spotify playlist songs and their corresponding audio features. Songs that matched songs in the Billboard dataset were removed before constructing the non-hit class. The remaining songs were assigned **hit = 0**. Songs from years beyond 2022 were also excluded so that the time range remained consistent with the Billboard dataset.

The main features used in the project include:

- Popularity
- Acousticness
- Danceability
- Energy
- Instrumentalness

- Liveness
- Loudness
- Speechiness
- Tempo
- Valence
- Key
- Mode
- Time Signature
- Song Length

The target variable is **hit**, representing whether the song belongs to the Billboard Hot 100.

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3. Approach

The project follows a supervised machine learning classification approach.

Data Preprocessing

First, both datasets were loaded and inspected for their structure and missing values. Song titles were normalized by converting them to lowercase, removing selected punctuation and standardizing whitespace.

The normalized song titles were used to identify songs already present in the Billboard dataset. Matching songs were removed from the Spotify playlist dataset so that a song would not incorrectly appear in both classes.

The corresponding Spotify features from the two datasets were then renamed into a common format and combined into a single dataset. Rows containing missing values were removed.

The resulting dataset was divided into input features **X** and target variable **y**. An **80:20 stratified train-test split** was used to maintain the class distribution in both sets.

Machine Learning Models

Three classification algorithms were implemented:

Logistic Regression:
Used as a baseline binary classification model to estimate the probability that a song belongs to the hit class.

Decision Tree:
Used to learn non-linear decision rules from the song features. A maximum depth was specified to control model complexity and reduce overfitting.

Random Forest:
Used as an ensemble model consisting of multiple decision trees. It was included to improve robustness and reduce the overfitting associated with an individual decision tree.

Feature scaling using **StandardScaler** was applied before model training.

4. Implementation and Results

The complete implementation was developed in **Python using Google Colab/Jupyter Notebook**. The project uses libraries including **Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn**.

The implementation performs the complete workflow:

Load Datasets

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Inspect & Clean Data

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Normalize Song Names

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Remove Billboard Matches

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Construct Hit / Non-Hit Labels

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Combine Datasets

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Train-Test Split

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Feature Scaling

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Train ML Models

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Evaluate Models

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Compare Results

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Feature Importance Analysis

The models were evaluated using **Accuracy, Precision, Recall and F1-score**. Confusion matrices were also generated to examine the classification results.

A model-comparison table and visualization were created to compare the three algorithms. Random Forest feature importance was also calculated to identify the Spotify features that contributed most to its predictions.

The final processed dataset was saved as **HitPredict_Final_Dataset.csv**, and the complete implementation is available in the project GitHub repository.

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5. Conclusion

The HitPredict project demonstrates how machine learning can be applied to music analytics to classify songs as Billboard hits or non-hits using Spotify-related features.

Logistic Regression, Decision Tree and Random Forest were implemented and evaluated using multiple classification metrics. The comparison helps identify the model that performs best on the constructed dataset, while feature-importance analysis provides insight into which song characteristics influence the predictions.

The project also has limitations. The non-hit class is constructed from Spotify playlist songs that were not matched with the Billboard dataset, so these songs cannot be guaranteed to have never appeared on Billboard. In addition, song matching is based primarily on normalized titles, and the random train-test split does not explicitly evaluate changes in musical trends over time.

Overall, the project provides a practical machine learning pipeline for exploring the relationship between Spotify song characteristics and Billboard hit classification.